

GUIDELINES FOR WRITING THE “DISCUSSION ON ALGORITHMS, MATHEMATICAL MODELS, AND FORMULAS.”

The discussion on **Algorithms, Mathematical Models, or Formulas** is a key section in the thesis for BSCS. Its purpose is to explain **what method you used, why it works, how it is designed, and how it performs**. This section allows your readers, including your panel, to understand the logic behind your solution and evaluate its performance. The discussion should be structured, clear, and systematic, covering the following subsections:

1. Fundamentals of the Algorithms/Mathematical Model/Formula

Start by introducing the **main concept** of your algorithm, model, or formula. This section should be **easy to understand**, even for someone unfamiliar with the specific method.

- Explain the **problem your algorithm or model is solving**.
- Define **inputs, outputs, variables, and parameters** clearly.
- Describe **how the method works in general terms**, without going into technical details yet.
- Provide **a small example or scenario** to illustrate the process.

Note: *Think of this as explaining your method to a peer. Avoid assuming your reader already understands the algorithm.*

2. Theoretical Foundations

Next, connect your algorithm or model to its **theoretical basis**. This subsection demonstrates that your method is scientifically grounded rather than merely a trial-and-error solution.

- Discuss **relevant concepts, theories, or principles** that support your method.
- Cite **previous studies or standard algorithms** that inspired your approach.
- Clearly state any **assumptions or limitations** of your method.
- Provide brief logical explanations if needed.

Note: *Use simple explanations, but make sure to show the logical reasoning behind your approach. This section builds credibility for your work.*

3. Algorithm Design

This is where you explain **how your algorithm is structured** and how it solves the problem step by step.

- Describe the **workflow of your algorithm** using words, flowcharts, or pseudocode.
- Define the **inputs, outputs, and intermediate steps**.
- Highlight any **decision-making processes, loops, or iterations**.
- Explain **why you chose this particular design** and how it fits the problem.

Note: *Even if your algorithm is simple, provide a clear explanation so that someone else could reproduce it. Visual diagrams help a lot.*

4. Mathematical Model Formulation (If Applicable)

If your study involves a **mathematical model**, you need to formalize it in this subsection.

- Define all **variables, constants, and parameters**.
- Present the **equations or formulas** that describe the system or process.
- State **constraints and boundaries**, if any.
- Include an **objective function** if the model is for optimization problems.

Note: *Your equations should clearly reflect how the algorithm or model works. This section shows the precision and logic behind your method.*

5. Algorithm Implementation

After explaining the design, describe how you **realized your algorithm in practice**.

- Mention the **programming language, tools, or libraries** you will use.
- Explain the **data structures or modules** that help the algorithm run efficiently.
- Use your collected dataset for machine learning algorithms.
- **Using 3rd party application is NOT allowed.**
- Describe **practical challenges** you encountered and how you solved them.
- **Show the output of implementing the algorithm.** Include **your own code snippets and model**, but keep it clear and readable.

Note: Focus on reproducibility. Another student should be able to implement your algorithm by following this section.

6. Complexity Analysis

Finally, evaluate your algorithm in terms of **efficiency and feasibility**.

- Discuss **time complexity** (how long it takes to run) and **space complexity** (memory usage).
- Compare your method with alternative approaches if relevant.
- Comment on **scalability** and whether the algorithm can handle larger datasets or real-world conditions.

Note: Even if your algorithm is simple, include this analysis to show that you considered its performance and limitations.

General Guidelines:

1. **Write clearly and systematically.** Each subsection should flow logically to the next.
2. **Use simple examples** to explain complex concepts.
3. **Include diagrams or pseudocode** wherever it helps understanding.
4. **Cite sources** if your algorithm or model is based on existing work.
5. **Be honest about limitations.** No algorithm is perfect, and acknowledging constraints is good research practice.